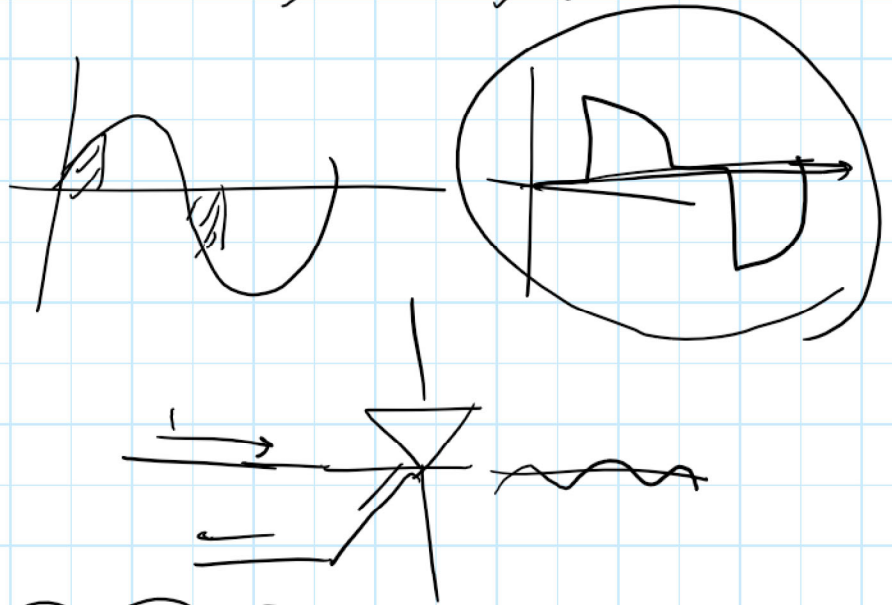
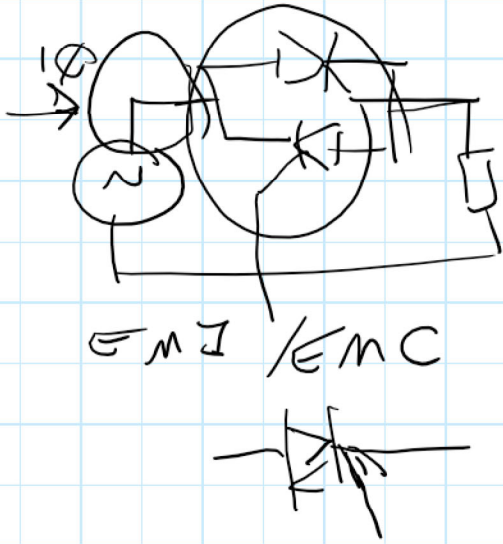


05-02-29

Tuesday, May 19, 2026 8:56 AM

Recall

AC/AC → Voltage Controllers



AC-voltage controller

$$V \propto \frac{V}{f} \propto B$$

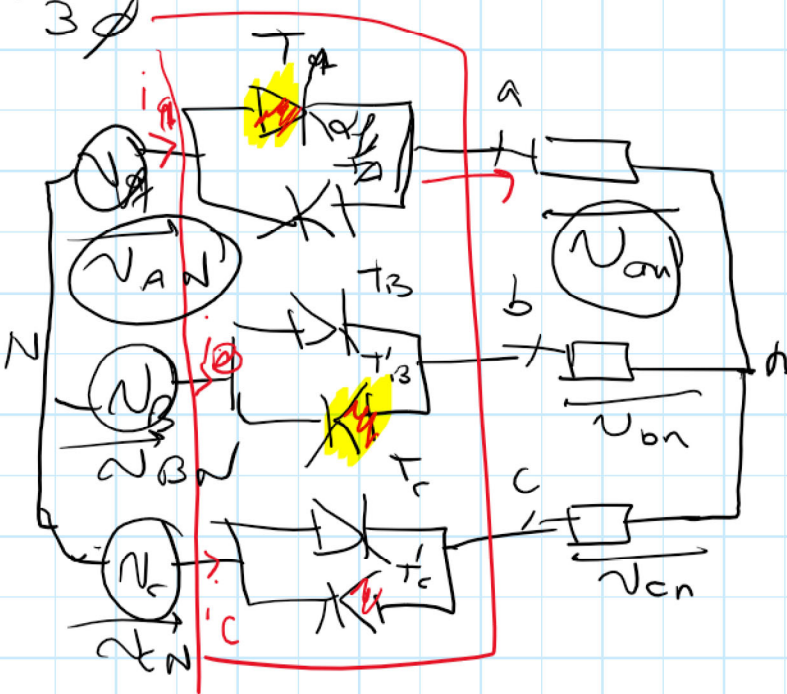
$$M = \frac{V_{an}}{V_{AN}} = ? (\neq 1 \neq \phi)$$

$$\phi \leq \alpha \leq 180^\circ$$

$$V_{an} = ?$$

$$\phi, \frac{1}{T_A}, \frac{1}{T'_A}$$

Maximum $\neq \phi$
SCRs conducting
= 3



$$T_A @ \phi \quad T'_A @ \phi + 180$$

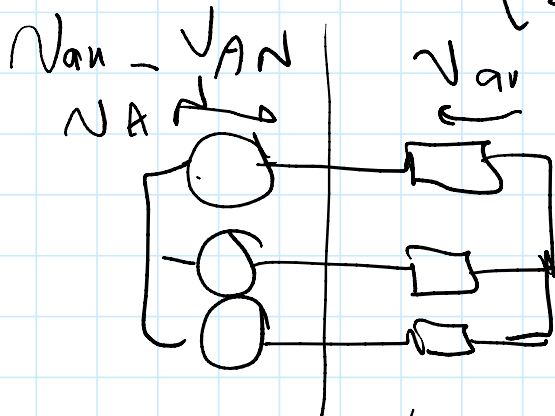
$$T_B @ \phi + 120 \quad T'_B @ \phi + 120 + 180$$

$$T_C @ \phi + 240 \quad T'_C @ \phi + 240 + 180$$

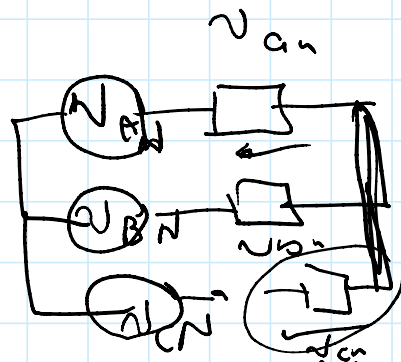
$T_C @ \phi = 240 \quad T'_C @ \phi = 240 + 180$ } SCR's on = 3

Minimum # of SCR's to have an output? 2

in each phase { NO SCR's is on $\Rightarrow V = \emptyset$ ✓
 3 SCR's are on $\Rightarrow V_o = V_A$ ✓
 2 SCR's are on $\Rightarrow V_o = \frac{1}{2} V_{AB}$

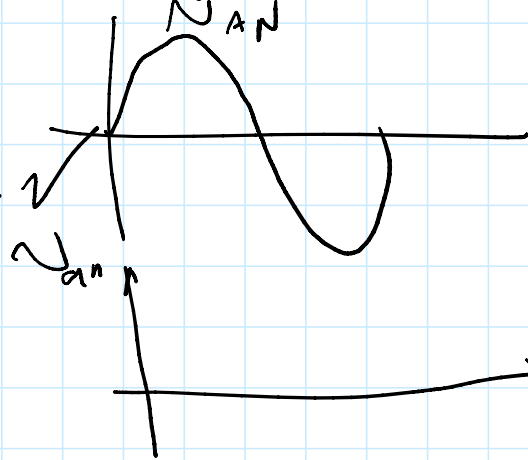


$V_{an} \begin{cases} \emptyset \\ V_{AN} \checkmark \\ \frac{1}{2} V_{AB} / \frac{1}{2} V_{AC} \end{cases}$



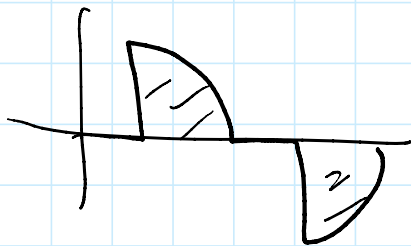
$V_{an} - V_{av}$
 $V_{AB} = V_{ab}$

$$V_{an} = \frac{1}{2} V_{AB}$$



$$\phi < \alpha_f < 180^\circ$$

in ϕ

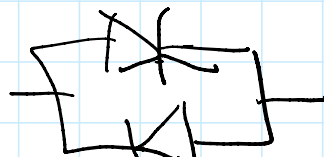


mode ①

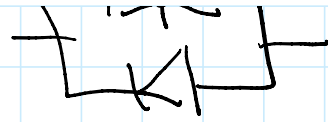
$$\phi < \alpha_f < 60^\circ$$

principles of operation

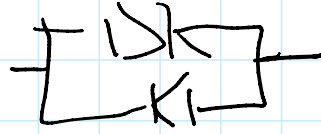
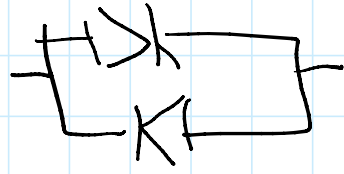
$$M = [\quad]$$



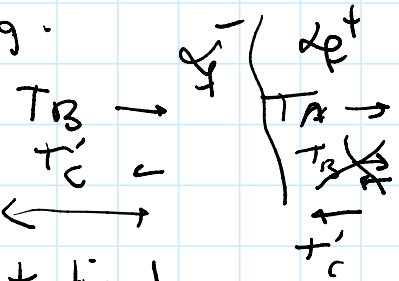
principles of operation
rules?



- ① starting from $\alpha_f(T_A)$
- ② assume that two SCRs were conducting @ α_f
- ③ check whether or not turning on T_A will turn one SCR off.



we must check the direction of current in 2 SCRs ^{that} were conducting: e.g.

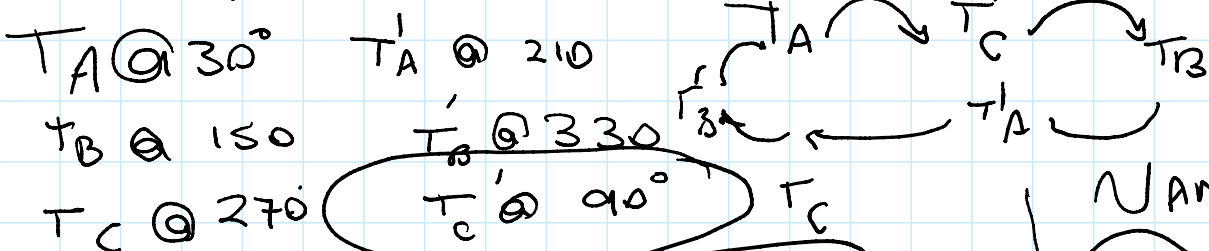


- ④ current zero-crossing must be checked (natural commutation)

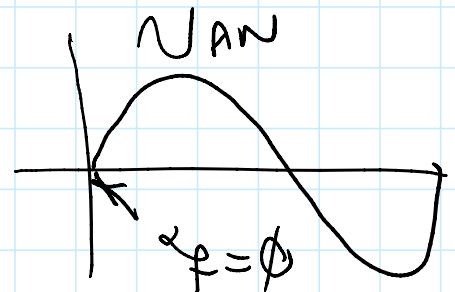
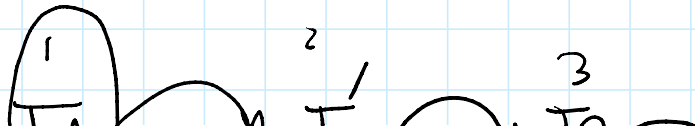
points to check $\rightarrow \alpha_f$ (firing angles)
to watch current zero crossing

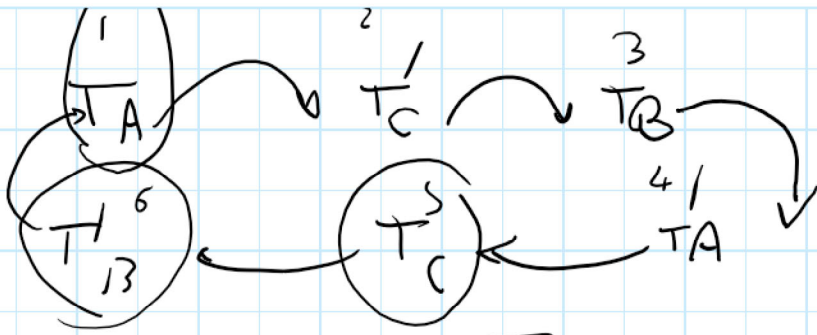
mode 1 if $\alpha_f < 60^\circ$ $N_{an} = \phi, V_{AN}, \frac{1}{2} V_{AC} / \frac{1}{2} V_{AB}$

e.g. $\alpha_f = 30^\circ$

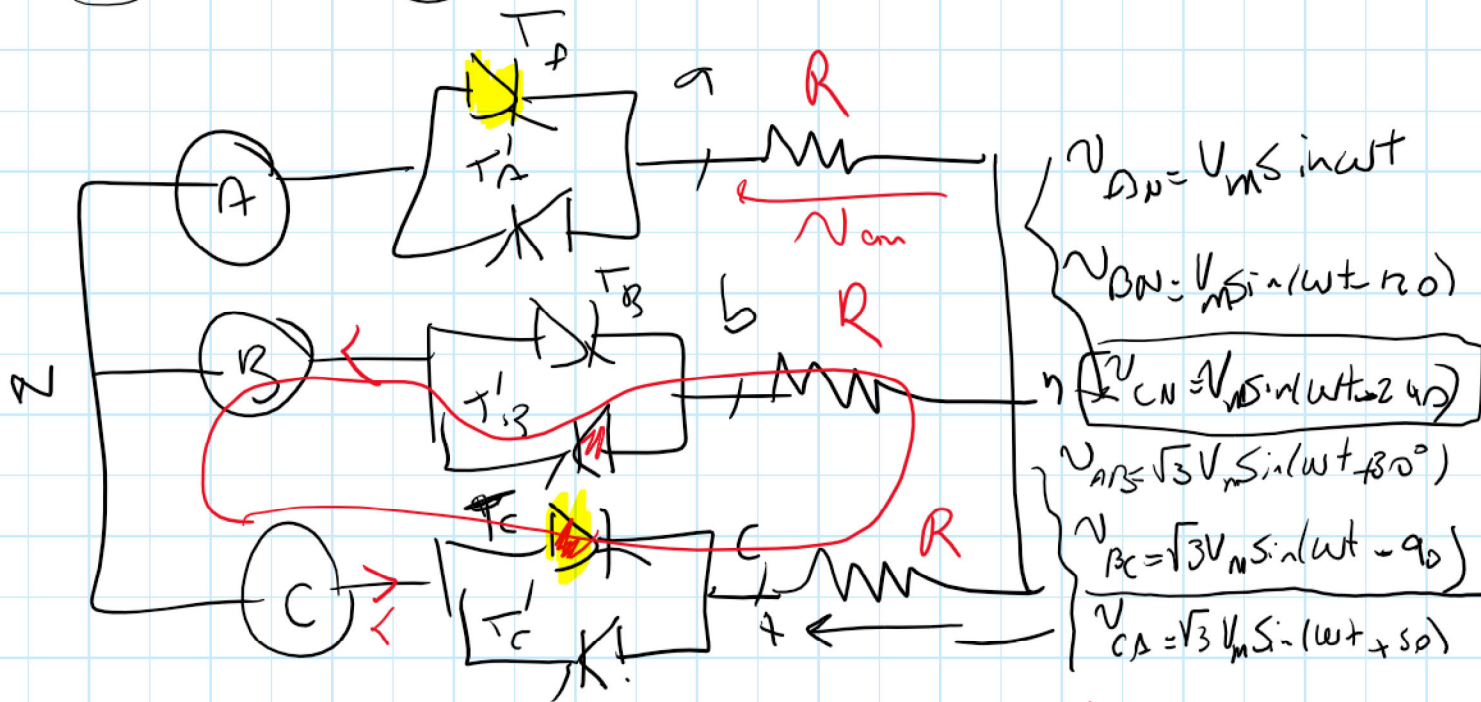


- ① T_A is gated @ $\omega t = \alpha_f = 30^\circ$





We assume a resistive load.



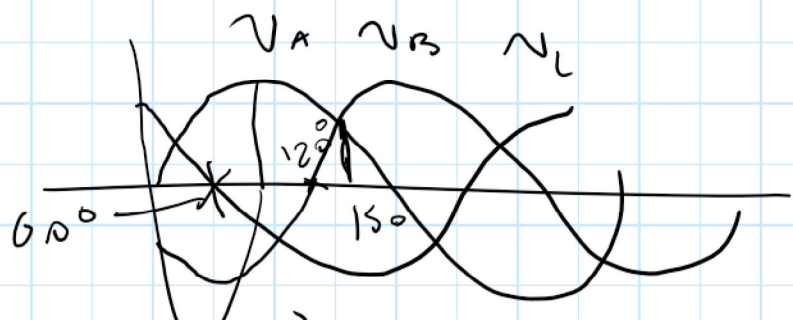
$\phi^- : 30^-$ ② $\phi = 30$ $\phi^+ = 30^+$

$I_C - V_{CN}$
 $V_{CN} = \frac{1}{2} V_{CB}$

$V_{CN} = V_{AN} > \phi$

② $T'_C @ 0^\circ$

$T_A \rightarrow V_{AN} = V_{AN}$
 $T'_B \leftarrow$



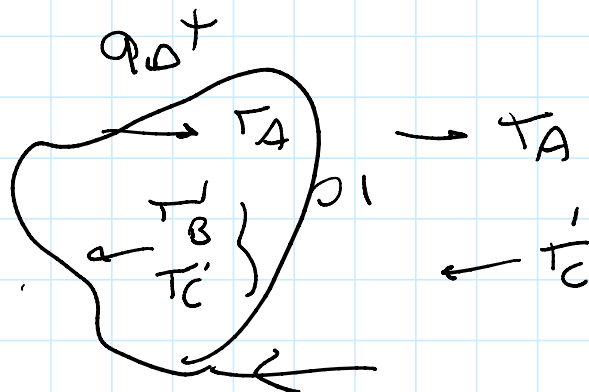
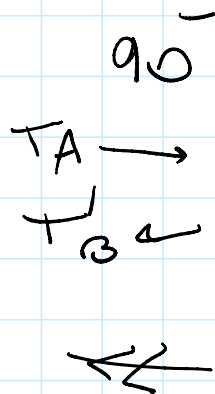
②

②

@ $\omega t = 0^\circ \Rightarrow V_{bn} = V_{cn} = 0 \Rightarrow i_b = 0 \Rightarrow$
 naturally commutator & turns off.

$T_A \rightarrow \quad V_{an} = \frac{1}{2} V_{AB}$
 $T'_B \leftarrow$

③ @ $\omega t = 90^\circ$, T'_c is gated.

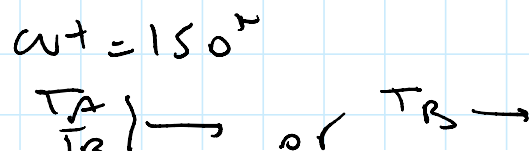
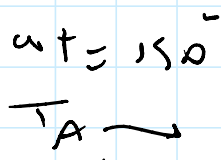


$T_A \rightarrow \quad V_{an} = V_{AN}$

④ @ 150° , @ $\omega t = 120^\circ$, $V_{bn} = V_{cn} = 0$

$V_{bn} = 0 \rightarrow$ resistive load $\rightarrow i_b = 0 \rightarrow T'_B$ naturally
 turns off. $T_A \rightarrow \quad T'_c \leftarrow \quad V_{an} = \frac{1}{2} V_{AC}$

⑤ @ $\omega t = 150^\circ$ T_B is gated.



$$\begin{array}{c} T_A \rightarrow \\ T'_C \leftarrow \end{array}$$

$$\begin{array}{c} T_A \\ T'_B \end{array} \rightarrow \text{ or } \begin{array}{c} T_B \rightarrow \\ T'_C \leftarrow \end{array}$$

① @ $\omega t = 210^\circ$

@ $\omega t = 180^\circ$ $V_{an} = V_{AN} = \phi \rightarrow i_a = \phi$

T_A turns off $\rightarrow V_{an} = \phi$